

On a new hyperchaotic system [☆]

Guoyuan Qi ^{a,b,*}, Michaël Antonie van Wyk ^a, Barend Jacobus van Wyk ^a, Guanrong Chen ^c

^a F'SATIE, Tshwane University of Technology, Pretoria 0001, South Africa

^b Department of Automation, Tianjin University of Science & Technology, Tianjin 300222, China

^c Department of Electronic Engineering, City University of Hong Kong, Kowloon, Hong Kong SAR, China

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Abstract

A new hyperchaotic system is presented, which has two large positive and one small negative Lyapunov exponent over a large range of parameters. As a consequence, system orbits strongly expand in some directions but rapidly shrink in some other directions. These strong expansions and shrinking lead the system orbits to be more disordered and random. Bifurcation and Poincaré-map analyses further show that the system has very rich bifurcations in different directions and extremely complicated dynamics overall. Spectral analysis shows that the system in the hyperchaotic mode has an extremely broad frequency bandwidth of high magnitudes, verifying its unusual random nature and indicating its great potential for some relevant engineering applications such as secure communications.

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1. Introduction

Hyperchaotic systems have more than one positive Lyapunov exponent (LE). Hyperchaos exists only in higher than or equal to four-dimensional (4D) autonomous systems. In general, hyperchaos is more disordered than ordinary chaos, and hyperchaotic systems have more complicated topological structures and dynamics than ordinary chaotic ones. Recently, hyperchaos has attracted increasing attention from various scientific and engineering communities. In secure communications, for example, a signal typically from a simple chaotic system was used to mask a message to be transmitted [1,2]. However, it is well known that an ordinary chaotic system

has the following disadvantages for such technological applications:

(1) It possesses only one single positive LE, so its degree of disorder is not high.

(2) Its high-magnitude bandwidth of the frequency spectrum is not very broad. For example, the bandwidth of chaotic signals generated numerically, not electronically, from the Lorenz system, the Rössler system [3] and the Chua's circuit [4] are all limited in between 0 and 4 Hz, which are too narrow to apply, therefore it cannot completely cover the masked messages in a real application. Once intercepted, there is a high possibility that the message can be extracted.

(3) Its orbits are periodic-like to some extent. When one observes the orbits of the Lorenz system, for instance, they travel between the two nonzero equilibria in a periodic-like fashion, which result in some undesirable properties such as narrow high-magnitude bandwidth and low-degree disorder.

Hyperchaos, in contrast, appears to be quite promising. Hyperchaos was first reported by Rössler [3], and the first circuit implementation of hyperchaos based on the Chua's circuit was realized by Matsumoto et al. [5]. Lately, there has been con-

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* Corresponding author at: Department of Automation, Tianjin University of Science & Technology, Tianjin 300222, China.

E-mail address: guoyuanqi@gmail.com (G. Qi).

siderable interest in the study of hyperchaos generation and its circuit implementation. For example, Nikolov and Clodong [6] modified the Rössler hyperchaotic system [3] by introducing linear feedback of its own signals; they also proposed a non-linear bi-directional coupling of two Chua's circuits [7]. By coupling three Chua's circuits together with sine functions in a ring, a new hyperchaotic attractor was generated by Cafagna and Grassi [8,9]. By modifying Chua's circuit, Thamilmaran et al. [10] presented a new hyperchaotic system. Kapitaniak et al. [11–13] have proposed two hyperchaotic systems by coupling five Chua's circuits, two Chua's circuits and two generalized van der Pol equations, respectively. The three-coupled Lorenz system was built by Reiterer et al. to realize hyperchaos via a triple Fourier series [14]. Recently, Li et al. obtained two hyperchaotic systems, by using a sinusoidal parameter perturbation to drive a unified chaotic system [15] and by using state feedback control [16], respectively. Hyperchaos was also found in coupled Colpitts oscillators [17]. Recently, hyperchaos synchronization has become an active research topic [18–20]. Last but not least, hyperchaos was also applied to devices, such as semiconductor lasers [6,21,22].

In general, the degrees of disorder of the trajectories of hyperchaotic systems are relatively high. However, there exist some common problems in most existing hyperchaotic systems. Firstly, two positive LEs simultaneously appear in a very narrow parameter range and their values are relatively small especially for the first positive LE [5–16]. Furthermore, the high-magnitude frequency bandwidths of these hyperchaotic signals are not broad enough for many engineering applications.

Recently, we proposed a 3D chaotic system [23] and a 4D chaotic system [24]. Some complicated dynamical behaviors, bifurcation analysis and circuit implementation of the 4D chaotic system were investigated and control [25,26]. We found that by using state-feedback, the two coexisting attractors generated by the 4D system can be merged to form a single complicated four-wing chaotic attractor [27]. We also found that the two new chaotic systems have broader frequency bandwidths of high magnitudes than the commonly known ordinary chaotic systems, even some commonly known hyperchaotic systems, as will be further discussed in Section 5 below. Furthermore, we found a new hyperchaotic system with complicated properties [29] based on the proposed 3D chaotic system [23].

In this Letter, we furthermore propose a new hyperchaotic system, and investigate its basic dynamics and properties. More precisely, we will analyze some complicated dynamics in detail by using LE spectra, bifurcation diagrams, Poincaré maps, and frequency spectral analysis. We will show that the system has two very large positive LEs over very wide parameter regions, so system orbits extensively expand in some directions but rapidly shrink in some other directions, which significantly increase the system's orbital degree of disorder and randomness. We will also show detailed bifurcation analysis, which illustrates the evolution processes of the system among sinks, periodic orbits, chaos and hyperchaos. Moreover, we will perform Poincaré-map analysis, which shows that the system has extremely rich dynamics. Finally, we will carry out frequency

spectral analysis, which demonstrates that the system has an extremely broad high-magnitude frequency bandwidth, very desirable for engineering applications such as secure communications.

2. The new 4D hyperchaotic system and its basic properties

Consider the following new 4D dynamical system:

$$\begin{aligned}\dot{x}_1 &= a(x_2 - x_1) + x_2x_3, \\ \dot{x}_2 &= b(x_1 + x_2) - x_1x_3, \\ \dot{x}_3 &= -cx_3 - ex_4 + x_1x_2, \\ \dot{x}_4 &= -dx_4 + fx_3 + x_1x_2.\end{aligned}\quad (1)$$

Here, x_i ($i = 1, 2, 3, 4$) are state variables and a, b, c, d, e, f are positive constant parameters.

2.1. Equilibria

To analyze the system, a good start is to find its equilibria, and then to characterize the local dynamical behaviors of the system orbits near these points based on the corresponding linearized systems. The spatial distribution and local (linearized) dynamical characteristics of the equilibria greatly influence the nonlinear dynamics of the system. The equilibria of system (1) can be found by solving the following algebraic equations simultaneously:

$$\begin{aligned}a(x_2 - x_1) + x_2x_3 &= 0, & b(x_1 + x_2) - x_1x_3 &= 0, \\ -cx_3 - ex_4 + x_1x_2 &= 0, & -dx_4 + fx_3 + x_1x_2 &= 0.\end{aligned}$$

Let

$$\begin{aligned}p &= \sqrt{a^2 + 6ab + b^2}, & g &= \frac{d - e}{cd + ef}, \\ q &= \sqrt{a^4 + a^2b^2 + 6a^3b}, & m_1 &= \frac{a^2 + 3ab + q}{b}, \\ m_2 &= \frac{a^2 + 3ab - q}{b}, & x_{11} &= \sqrt{\frac{2m_1}{g}} \frac{a}{(2a - m_1)}, \\ x_{12} &= \sqrt{\frac{2m_2}{g}} \frac{a}{(2a - m_2)}, & x_{21} &= \sqrt{\frac{m_1}{2g}}, \\ x_{22} &= \sqrt{\frac{m_2}{2g}}, & x_{31} &= gx_{11}x_{21}, & x_{32} &= gx_{12}x_{22}, \\ x_{41} &= \frac{c + f}{cd + ef} x_{11}x_{21}, & x_{42} &= \frac{c + f}{cd + ef} x_{12}x_{22}.\end{aligned}$$

It is easily to prove that $q, m_1, m_2 > 0$, $2a - m_1 \neq 0$ and $2a - m_2 \neq 0$, when the parameters a, b, c, d, e, f are positive constants.

Remark 1. When $d < e$, i.e. $g < 0$, system (1) has only one equilibrium, which is the origin $S_0 = [0, 0, 0, 0]^T$; when $d > e$, i.e. $g > 0$, system (1) has 5 real equilibria, which are denoted by

$$S_0, \quad S_1 = [x_{11}, x_{21}, x_{31}, x_{41}]^T,$$

$$\begin{aligned} S_2 &= [-x_{11}, -x_{21}, x_{31}, x_{41}]^T, \\ S_3 &= [x_{12}, x_{22}, x_{32}, x_{42}]^T, \quad S_4 = [-x_{12}, -x_{22}, x_{32}, x_{42}]^T. \end{aligned} \quad (2)$$

By linearizing system (1) at S_0 , one obtains the Jacobian

$$A_0 = \begin{bmatrix} -a & a & 0 & 0 \\ b & b & 0 & 0 \\ 0 & 0 & -c & -e \\ 0 & 0 & f & -d \end{bmatrix}. \quad (3)$$

The eigenvalues of matrix A_0 are

$$\begin{aligned} \lambda_{01} &= (b - a - p)/2, & \lambda_{02} &= (b - a + p)/2, \\ \lambda_{03} &= -(c + d - r)/2, & \lambda_{04} &= -(c + d + r)/2, \end{aligned} \quad (4)$$

where $r = \sqrt{c^2 - 2cd + d^2 - 4ef}$. And their corresponding eigenvectors are

$$\begin{aligned} v_{01} &= [-(b + a - p)/(2b), 1, 0, 0]^T, \\ v_{02} &= [-(b + a + p)/(2b), 1, 0, 0]^T, \\ v_{03} &= [0, 0, (d - c + r)/(2f), 1]^T, \\ v_{04} &= [0, 0, (d - c - r)/(2f), 1]^T. \end{aligned} \quad (5)$$

Since a, b, c, d, e, f are all positive numbers, $\lambda_{02} > 0$, and the real parts of the other eigenvalues $\lambda_{01}, \lambda_{03}, \lambda_{04}$ are all negative, implying that the equilibrium S_0 is always a saddle point. The linearized stable and unstable subspaces of S_0 are expressed as

$$E^s(S_0) = \text{span}\{v_{01}, v_{03}, v_{04}\}, \quad E^u(S_0) = \text{span}\{v_{02}\}. \quad (6)$$

2.2. Symmetry and similarity

System (1) has certain symmetry and similarity, as summarized below.

Remark 2. System (1) is symmetric with respect to the coordinate plane x_3 – x_4 , which is easily proved via the following transformation:

$$(x_1, x_2, x_3, x_4) \rightarrow (-x_1, -x_2, x_3, x_4). \quad (7)$$

Equilibria S_1, S_2 are symmetric with respect to the plane x_3 – x_4 . The same for equilibria S_3, S_4 .

Remark 3. Equilibria S_1, S_2 , are similar to one another in the sense that they have the same group of eigenvalues; the second kind of equilibria, S_3, S_4 , are also similar to one another, and have another same group of eigenvalues.

To verify the above, let A_i denote the Jacobians of S_i , $i = 1, 2, 3, 4$. Then, one has

$$A_1 = \begin{bmatrix} -a & a + x_{31} & x_{21} & 0 \\ b - x_{31} & b & -x_{11} & 0 \\ x_{21} & x_{11} & -c & -e \\ x_{21} & x_{11} & f & -d \end{bmatrix}, \quad (8.1)$$

$$A_2 = \begin{bmatrix} -a & a + x_{31} & -x_{21} & 0 \\ b - x_{31} & b & x_{11} & 0 \\ -x_{21} & -x_{11} & -c & -e \\ -x_{21} & -x_{11} & f & -d \end{bmatrix}, \quad (8.2)$$

$$A_3 = \begin{bmatrix} -a & a + x_{32} & x_{22} & 0 \\ b - x_{32} & b & -x_{12} & 0 \\ x_{22} & x_{12} & -c & -e \\ x_{22} & x_{12} & f & -d \end{bmatrix}, \quad (9.1)$$

$$A_4 = \begin{bmatrix} -a & a + x_{32} & -x_{22} & 0 \\ b - x_{32} & b & x_{12} & 0 \\ -x_{22} & -x_{12} & -c & -e \\ -x_{22} & -x_{12} & f & -d \end{bmatrix}. \quad (9.2)$$

It can be verified that there exists a transformation matrix D_{12} between A_1 and A_2 , such that

$$D_{12}^{-1} A_1 D_{12} = A_2, \quad D_{12}^{-1} A_3 D_{12} = A_4, \quad (10)$$

where $D_{12} = \text{diag}(-1, -1, 1, 1)$, and D_{12} is an orthogonal matrix, since

$$D_{12}^{-1} = D_{12}^T = D_{12}. \quad (11)$$

Next, let $V_i = [v_{i1}, v_{i2}, v_{i3}, v_{i4}]$ denote the matrix composing of eigenvectors of S_i , $i = 1, 2, 3, 4$. Then, the following identities can be easily obtained:

$$D_{12}^{-1} V_1 D_{12} = V_2, \quad D_{12}^{-1} V_3 D_{12} = V_4. \quad (12)$$

Also, let $W^s(S_i)$, $W^u(S_i)$, $W^c(S_i)$ denote the stable, unstable, and center manifolds of S_i , $i = 1, 2, 3, 4$, respectively.

Definition 1. The two similar and symmetric equilibria are called as an equilibrium-pair.

Remark 4. $S_{1,2}$ and $S_{3,4}$ are two distinct equilibrium-pairs, and every equilibrium-pair has the same local stable, unstable and center manifolds.

3. Observation and analysis of the new hyperchaotic attractor

When one solves the Lorenz system by using the Runge–Kutta method, a sampling time of 10^{-2} (s) is feasible to obtain a smooth trajectory. However, system (1) is extremely sensitive to the sampling time. The time step must be adequately small, such as $\tau = 2 \times 10^{-4}$ (s), or $\tau = 1 \times 10^{-4}$ (s) for some parameters, due to the rapid change of the hyperchaotic orbits of the system.

3.1. Analysis of Lyapunov spectra

Firstly, the LE spectra and Lyapunov dimension (LD) [28] are studied. Let $b \in [15.425, 27]$, the parameter interval be 0.025, and the running time be $T = 500$ (s) at each parameter point, so $n = T/\tau = 2.5 \times 10^6$ steps are calculated for each of the four LEs. LD is then calculated according to the following criterion: If there is no positive LE, then the LD is zero; if there are positive Lyapunov exponents, then LD is $l_d = j + \sum_{i=1}^j l_i / |l_{j+1}|$, where j is such that $\sum_{i=1}^j l_i \geq 0$ and $\sum_{i=1}^{j+1} l_i \leq 0$.

Fig. 1 indicates the LE spectra with $b \in [15.425, 27]$. One can observe that there are two positive LEs over quite a wide

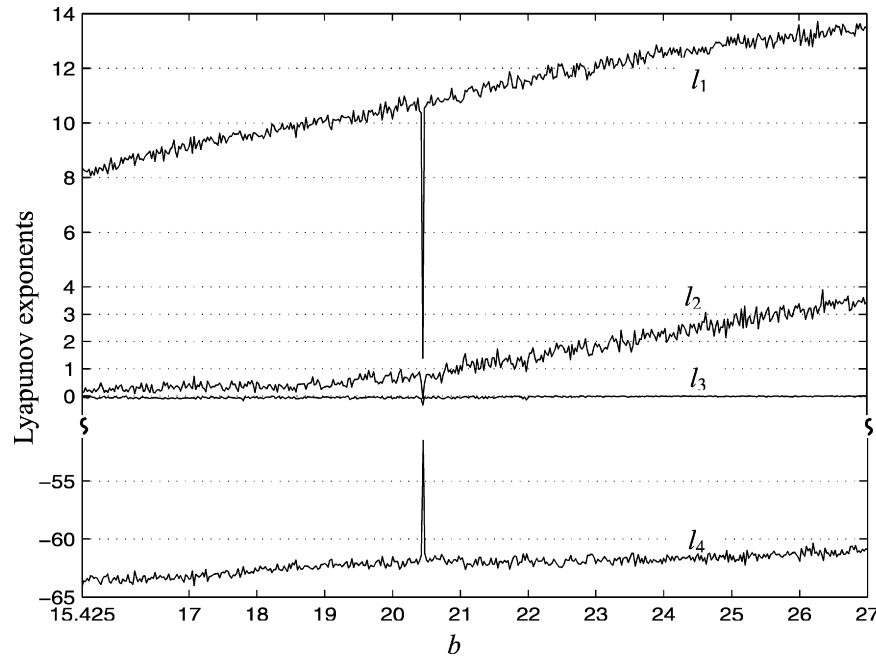


Fig. 1. The Lyapunov exponent spectra versus b , with $a = 50c = 13$, $d = 8$, $e = 33$, $f = 30$.

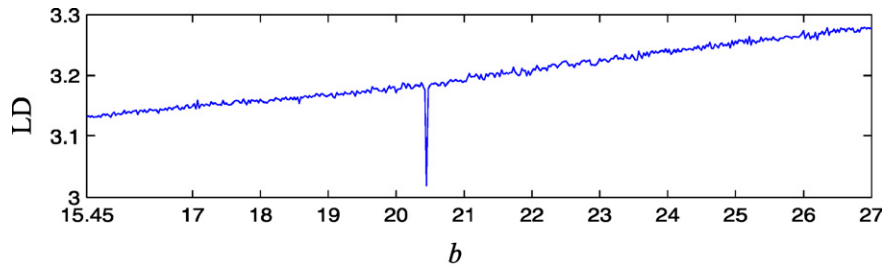


Fig. 2. The Lyapunov dimension versus b , with $a = 50c = 13$, $d = 8$, $e = 33$, $f = 30$.

range of parameters, which implies that the system is hyperchaotic over a broad range. The first LE is very large with $l_1 \in [8.3585, 13.4632]$, the second LE is considerable large with $l_2 \in [0.1, 3.4781]$, the third LE is approximately zero, and the forth LE is negative with $l_4 < -60$. As will be clearly seen below, when the parameter b is varying, the second LE will become even more prominently positive. Fig. 2 shows the LD spectrum. It is noticeable that the system has only one equilibrium, i.e. the origin, under these parameters setting according to Remark 1.

The leading LE is less than 2 for most existing hyperchaotic systems. For example, they are about 0.11 [3], 0.24 [5], 0.2 [6], 0.35 [7], 0.35 [8], 0.77 [9], 0.1 [10], 0.4 [11–14], 0.6 [16], and 1.7 [15], respectively. The second largest LE is also relatively small with $l_2 \in [0.02, 0.18]$ [3,5–8,10,12–16]. However, the leading LE l_1 of the new hyperchaotic system here is notably equal to 13.4632. Similarly, the second largest LE is considerably large and positive, with a value of 3.4781. The exponential expansions of an attractor with positive LEs are incompatible with motions on a bounded space unless there are many times of foldings. Obviously, the values of the leading LE and the second LE imply an exponential rate of stretching and folding, i.e. $e^{l_1 t}$ and $e^{l_2 t}$, in two different directions, as the orbit moves.

This can be a good measure as how chaotic the attractor is. For comparison, consider a hyperchaotic system with two positive LEs, $l_1 = 0.6$, $l_2 = 0.1$. Then, the rates of stretching and folding of the new hyperchaotic system, which has $l_1 = 13.4632$ and $l_2 = 3.4781$, will be n times and m times faster than that of the assumed hyperchaotic system, where

$$\begin{aligned} n &= \frac{e^{13.4632t}}{e^{0.6t}} \Big|_{t=1} = 2.4280 \times 10^5, \\ m &= \frac{e^{3.4781t}}{e^{0.1t}} \Big|_{t=1} = 29.3121. \end{aligned} \quad (13)$$

That explains why system (1) is so sensitive to the sampling time (with a very small $\tau = 1 \times 10^{-4}$). This also implies that the system has extremely rich frequency properties, as will be further proved by the more detailed frequency analysis given in Section 5.

3.2. Observation in different views

Fix the parameter $b = 24$. Then, the LEs are $[l_1, l_2, l_3, l_4] = [12.3981, 2.2944, -0.0009, -61.6578]$, and the LD is 3.2381. Fig. 3(a)–(j) show the system orbits, displayed in different

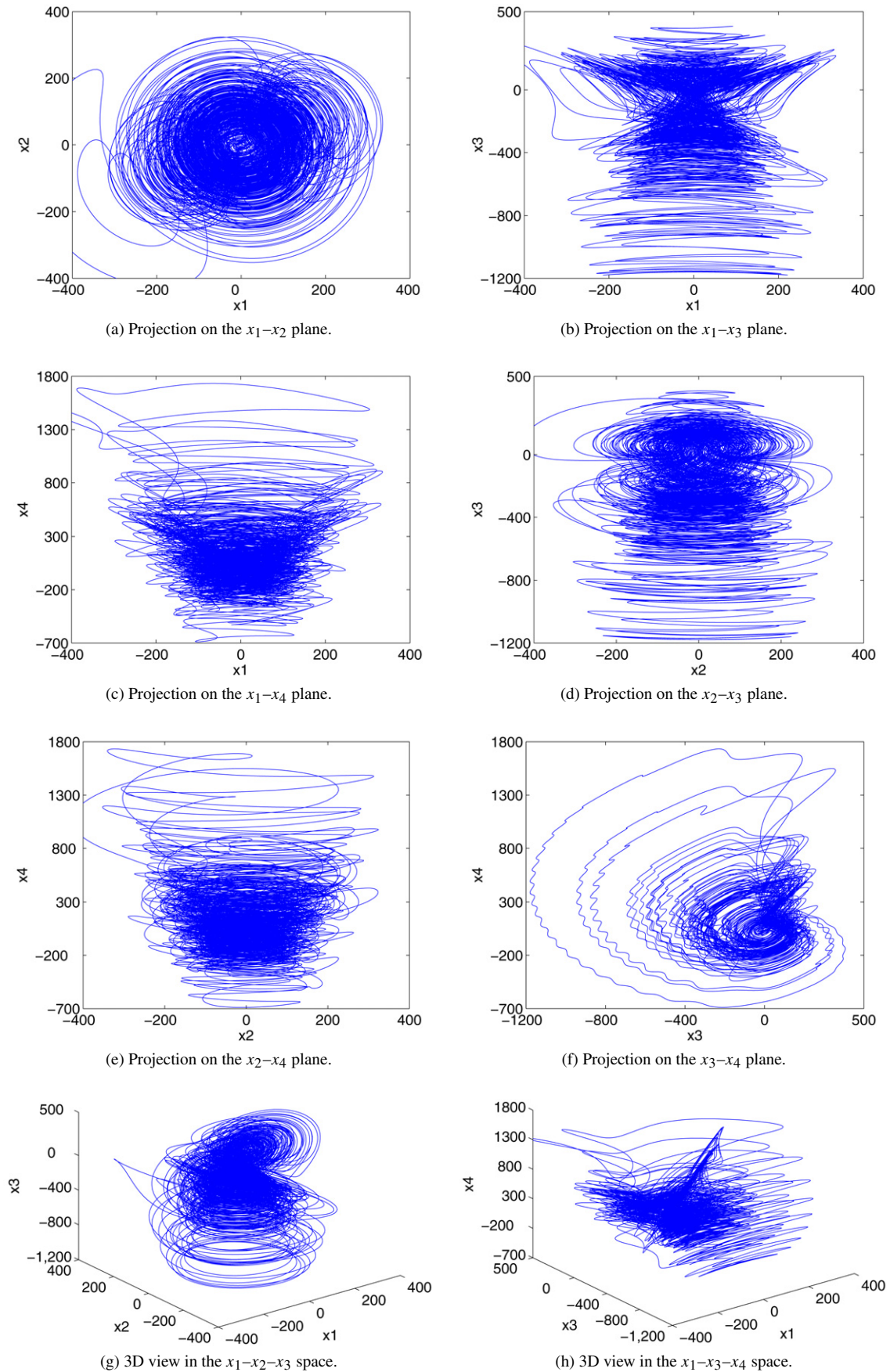


Fig. 3. The orbits of system (1) with parameters $a = 50$, $b = 24$, $c = 13$, $d = 8$, $e = 33$, $f = 30$.

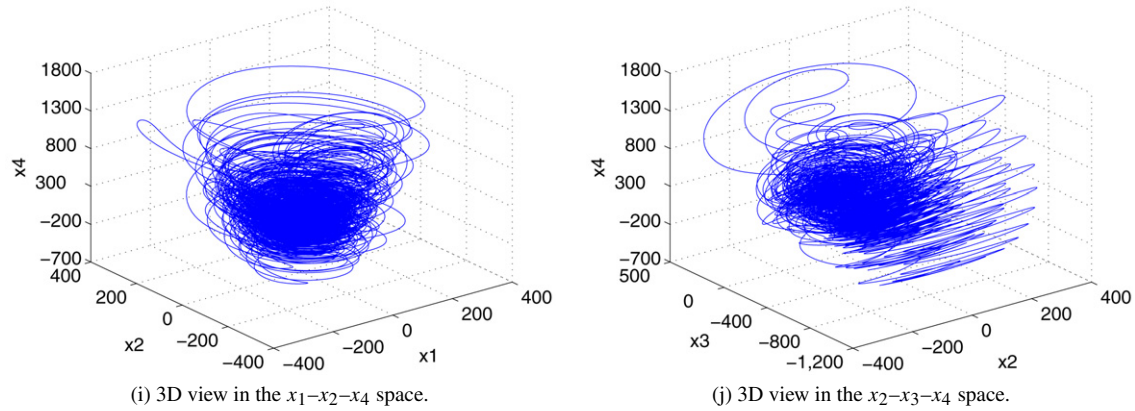


Fig. 3. (continued)

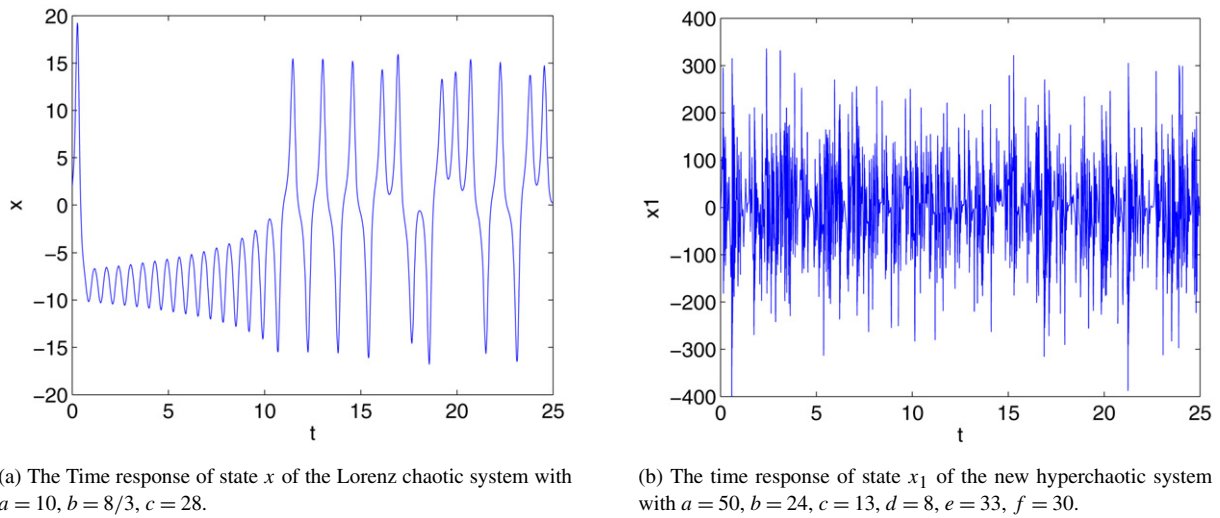


Fig. 4. The time responses of the Lorenz system and the new hyperchaotic system.

spaces, where the running time is 25 (s). It can be seen that:

- (1) the orbits are more complicated and disordered than the ordinary chaotic and hyperchaotic ones;
- (2) the attractors have very irregular forms, neither butterfly nor scroll ones, which are visually more complex than the ordinary chaotic attractors;
- (3) the orbits are extremely abundant and dense as compared with the orbit of the Lorenz system, which takes on a relatively regular and simple form, as shown in Figs. 4(a), (b) with the same running time $T = 25$ (s).

3.3. Pseudo-randomness of the hyperchaotic system

A histogram can statistically show the stochastic properties of a signal generated by a system. Figs. 5(a), (b) are the estimated probability densities of the Lorenz system and system (1), respectively. It is clear that the stochastic properties of the hyperchaotic orbits are very similar to that of Gaussian white noises, but that of the Lorenz system has no such property. The bins of histogram are $Y = 44\,000$, and $M = 2000$ for both systems.

A basic and prominent property of a chaotic system is its seemingly erratic behavior, where a key element is the sen-

sitivity of the trajectory to the initial conditions. Trajectories are usually predictable for a short period of time. But, over a long period, chaotic trajectories become unpredictable. Different chaotic systems have different degrees of sensitivity to initial conditions, which differentiate their degrees of disorder. Figs. 6(a) and (b) show the time responses from two initial conditions of the hyperchaotic system in [16] and of the new hyperchaotic system (1), respectively. For clarity here, the period is taken to be only 1.5 (s); otherwise, the time response of the new hyperchaotic system will be too dense to view. Here, the solid lines and the dotted ones denote the trajectories at two initial conditions from the two hyperchaotic systems, respectively. The orbit of the new hyperchaotic system shows a stronger sensitivity to initial conditions and is more unpredictable over the same period of time than those of the hyperchaotic system in [30], the reason being that the distance of the trajectories from two initial conditions increase exponentially on average. The rates of divergence are measured by the positive LEs. The positive LEs of the hyperchaotic system in [30] are $l_1 = 0.5011$ and $l_2 = 0.1858$, respectively; but they are much bigger, $l_1 = 13.4632$ and $l_2 = 3.478$, in the new hyperchaotic system (1).

In fact, the differential state vector $\dot{x} = [\dot{x}_1, \dot{x}_2, \dot{x}_3, \dot{x}_4]$ of the orbit represents its rate of change, change of direction, and de-

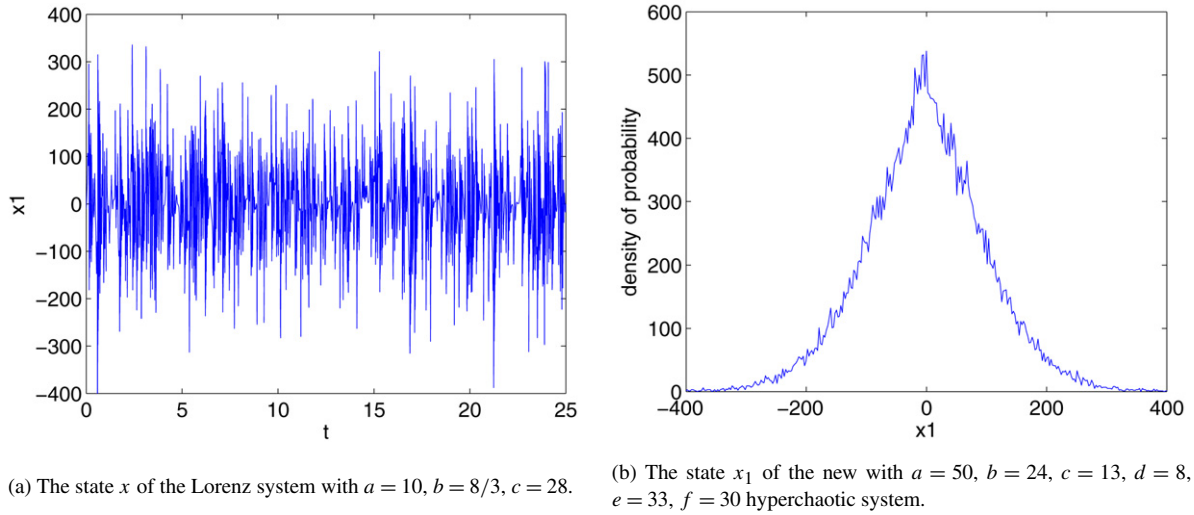


Fig. 5. A comparison of the probability densities between the Lorenz system and the new hyperchaotic system.

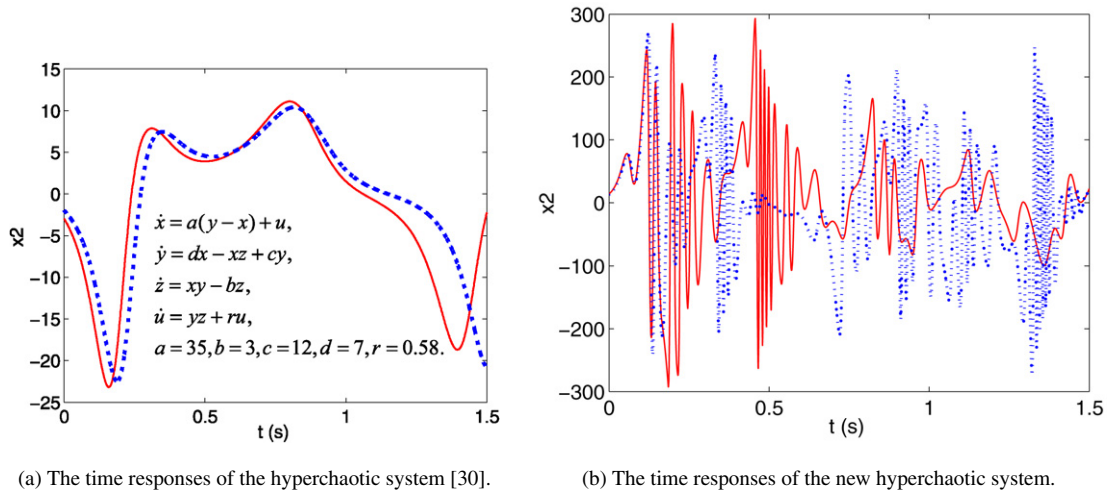


Fig. 6. A comparison of sensitivities to initial conditions for two hyperchaotic systems, with $a = 50$, $b = 26$, $c = 13$, $d = 8$, $e = 33$, $f = 30$.

gree of smoothness. Larger differential values indicate faster and more drastic changes of system orbital motions, and more directions mean richer folding behaviors and more complicated bifurcations of the orbits.

The differential vector $[\dot{x}_1, \dot{x}_2, \dot{x}_3, \dot{x}_4]$ can be easily calculated from system (1), as shown in Fig. 7(a). Obviously, the rate of change $\dot{x}_i \in [-10^5, 10^5]$, $i = 1, \dots, 4$, and the change of direction, are spreading over the whole 4D space, indicating that the changing pattern is extremely complex. For clarity, Fig. 7(a) is magnified by 20×20 times about $\dot{x}_1$ – $\dot{x}_3$ and then shown in Fig. 7(b), where the orbits are given by ‘.’, which reflects the density of the orbital points and the randomness of the change of directions in the plane, where one can clearly see the tortuous curves of the motion velocity.

3.4. The Poincaré map

As an important technique, the Poincaré map is used to describe bifurcation and folding properties of chaos. Take $\Sigma = \{(x_i, i = 1, 2, 3, 4) \in R^4 \mid x_1 = 0\}$ as a crossing section. The associated Poincaré map is the mapping on Σ from the 4D space to the 3D subspace x_2 – x_3 – x_4 , as shown in Fig. 8(a).

Figs. 8(b)–(d) are projections of the Poincaré map on different phase planes. The 3D map looks like two jointed large and thickset trees with a number of branches, implying that the orbit is continuously folded and bifurcated in various directions. It is noticeable that the Poincaré maps of many known chaotic systems only show a slim branch with several twigs on 2D surfaces; however, one can see from Figs. 8(b)–(d) that the Poincaré map here consists of several large limbs with various bifurcations in different directions, which indicates that the system has extremely rich dynamics. The distinction basically lies in that an ordinary chaotic system only has one small positive LE, so the system orbit expanding in one direction; while this hyperchaotic system has two large positive LEs, corresponding to the fact that the orbit is drastically expanding in two directions.

3.5. The bifurcation analysis with $a = 50$, $c = 13$, $d = 8$, $e = 33$, $f = 30$, $b \in [0, 15]$

Figs. 9 and 10 show the LEs spectrum and bifurcation diagram with $b \in [0, 15]$, respectively. Both the LEs spectrum and the bifurcation diagram demonstrate that the system can

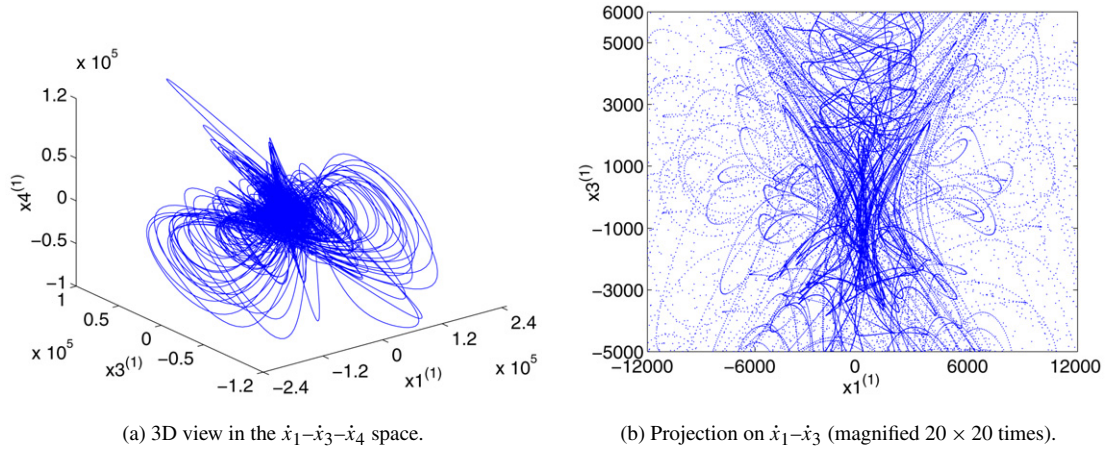


Fig. 7. Velocity of the orbital motion of system (1) with parameters setting: $a = 50$, $b = 24$, $c = 13$, $d = 8$, $e = 33$, $f = 30$.

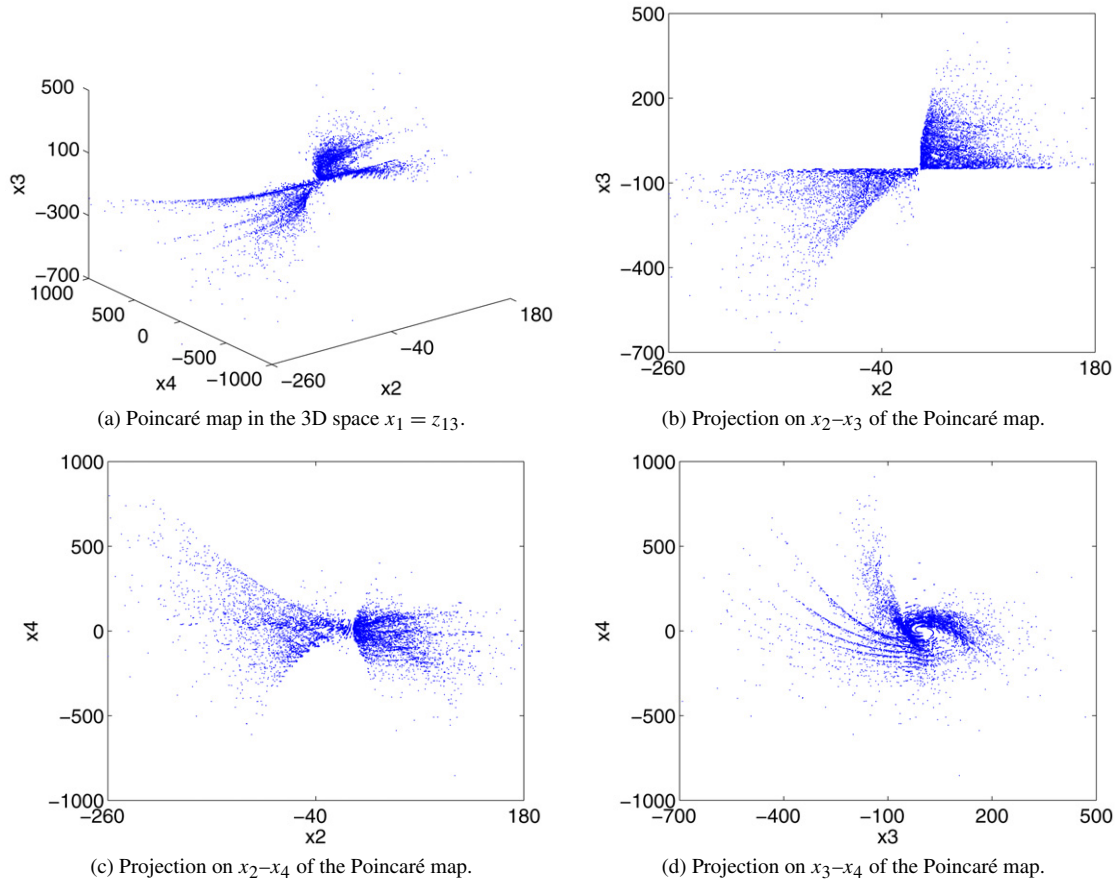


Fig. 8. Poincaré map of system (1) with $a = 50$, $b = 24$, $c = 13$, $d = 8$, $e = 33$, $f = 30$, $x_1 = 0$.

generate complex dynamics within a wide range of parameters, including different kinds of complex periodic orbits, bifurcations and chaos, etc. One can observe very complicated periodic orbits from Figs. 11(g) and (h). The evolution processes in different modes, corresponding to the different parameter regions for b , are described in more details in Table 1 and Fig. 11. It is noticeable that the bigger the leading LE, the more complex the chaos, as shown in Fig. 11(d) with $LE = 1.496$, Fig. 11(f) with $LE = 1.66$, Fig. 11(h) with $LE = 1.771$, Fig. 11(i) with $LE = 3.18$, and Fig. 11(k) with $LE = 6.44$, respectively.

As parameter b increases, the first LE becomes bigger and bigger, the second LE still keeps zero. When $15 < a < 15.45$,

the first LE continuously increases and $l_1 > 8$. The second LE is greater zero, but it is relatively small, i.e. $0 < l_2 < 0.1$, which is coincident with that the shape of the chaotic attractor is similar to that of the hyperchaotic attractor in the parameter region. There is no dominant difference in transition process from the chaotic attractor to the hyperchaotic attractor.

4. Frequency spectral analysis

Many proposed chaos-based encryption schemes have been totally or partially broken by different attacks. One reason is that the degree of randomness of simple chaotic or hyperchaotic

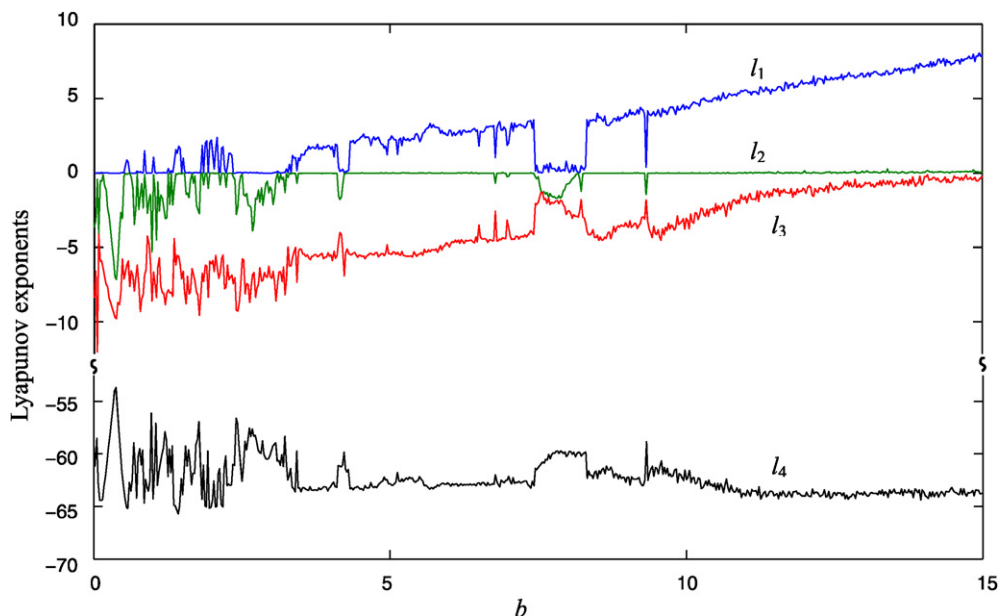


Fig. 9. The LE spectra of system (1) versus b , with $a = 50$, $c = 13$, $d = 8$, $e = 33$, $f = 30$.

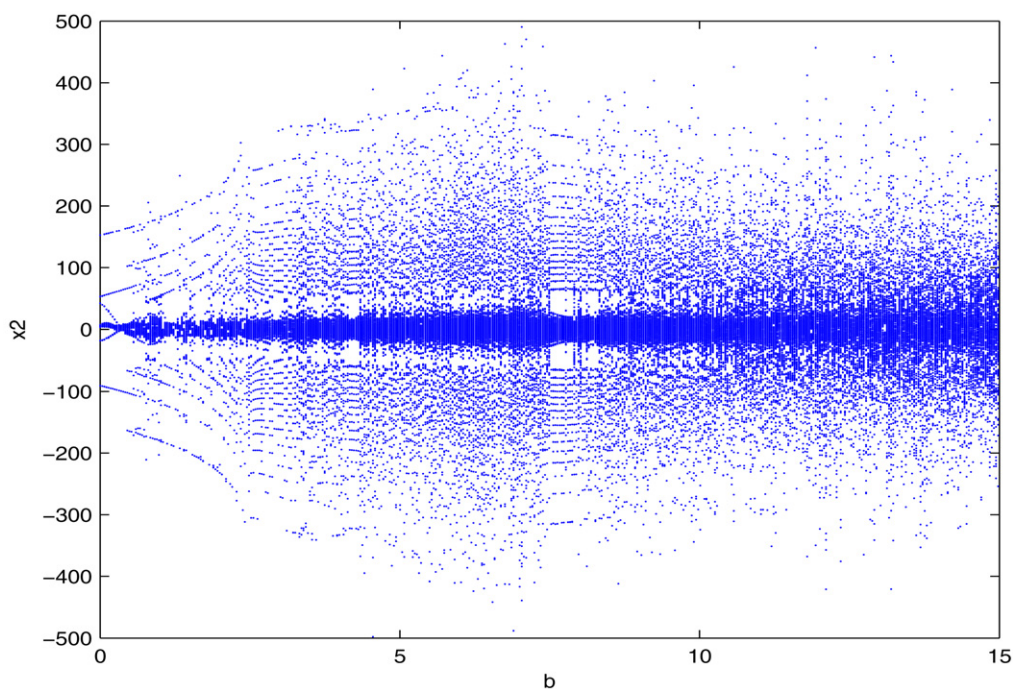


Fig. 10. Bifurcation diagrams of x_3 with $a = 50$, $c = 13$, $d = 8$, $e = 33$, $f = 30$.

signals used therein are not high enough, as reflected by the narrow high-magnitude bandwidths of those signals [30–32].

Fig. 12(a) shows the frequency spectra of the Lorenz system. For comparison, the Runge–Kutta method was used to solve all systems, all with sampling time 0.0002 (s), running time 0–500 (s), number of spectral averages 100, and all spectra are normalized, where only the frequency ranges with spectral values over 10^{-1} are considered. The high-magnitude bandwidths of the Lorenz system, the Chua circuit and the Rössler hyperchaotic system are all less than 4 Hz numerically. In practice, the frequency of the analog signal of a chaotic system can be

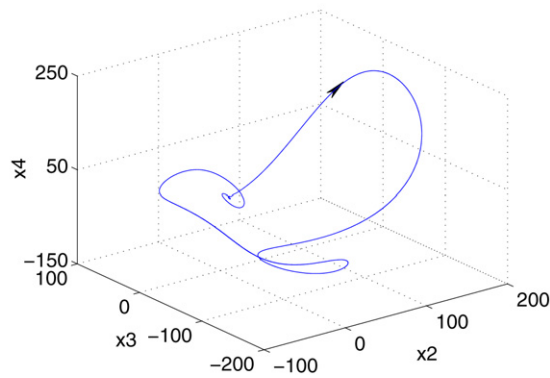
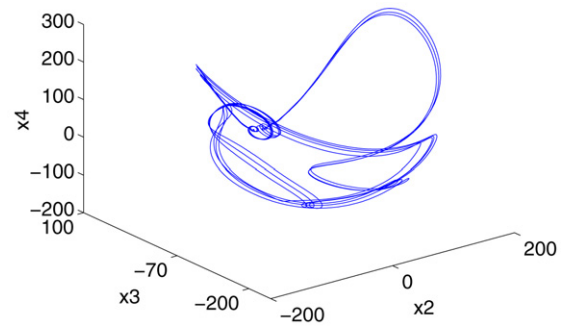
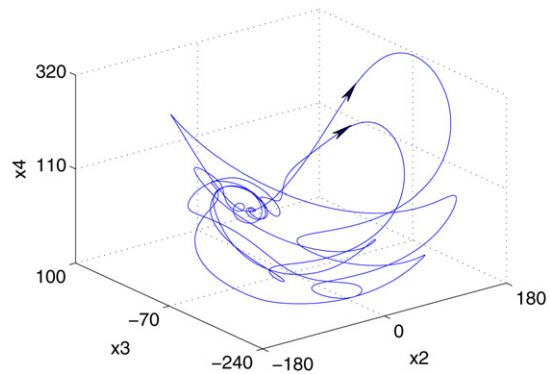
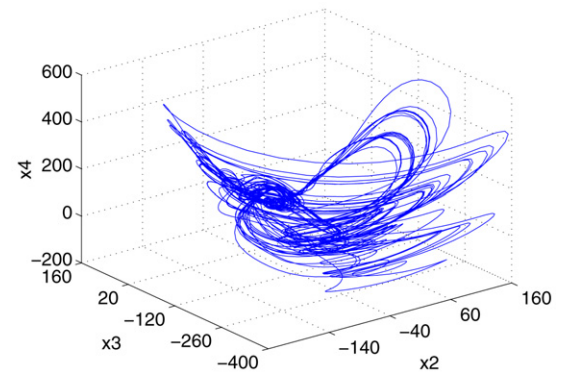
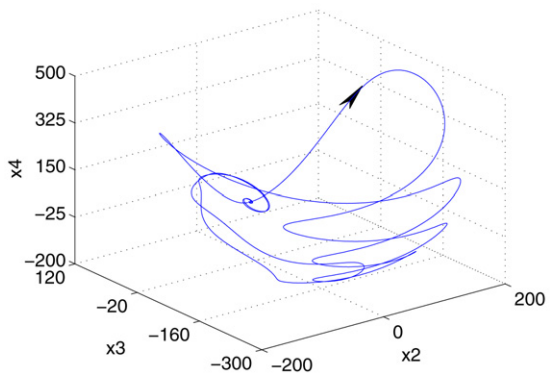
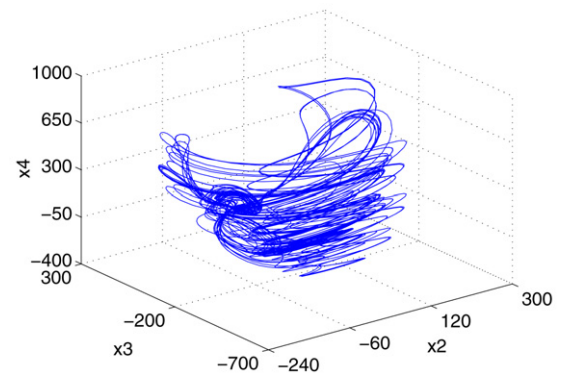
extended to several kHz, even several MHz, therefore 4 Hz is essentially useless. To compare, only the bandwidths of the signals generated numerically by the system itself are discussed here, without investigating possible electronic techniques that may somewhat improve the bandwidths of such chaotic systems including the new system.

Similarly, the high-magnitude spectral bandwidths of the Rössler system and the Chua circuit are both less than 4 Hz. Therefore, the chaotic signals cannot be used to sufficiently mask the messages in real communication applications. Once intercepted, there is a high possibility that the messages can be

Table 1

Align the orbital modes with respect to b with $a = 50$, $c = 13$, $d = 8$, $e = 33$, $f = 30$

Regions of parameter b	1st LE	Orbital modes	Figures of modes
< 0.5	Zero	Periodic orbit	Fig. 11(a)
$[0.5, 0.6)$	Positive	Multi period-doubling orbits	Fig. 11(b)
$[0.6, 0.825] \cup [0.875, 0.975)$	Zero	Period-doubling orbits	Fig. 11(c)
$(0.825, 0.875) \cup [0.975, 1.025] \cup [1.35, 1.525]$	Positive	Chaos	Fig. 11(d)
$(1.025, 1.35) \cup (1.525, 1.8]$	Zero	Complex periodic orbit	Fig. 11(e)
$(1.8, 2.35)$	Positive	Chaos	Fig. 11(f)
$[2.35, 3.275] \cup [4.125, 4.3]$	Zero	Periodic orbit	Fig. 11(g, h)
$(3.275, 4.125)$	Positive	Chaos	Fig. 11(i)
$(4.3, 7.45)$	Positive	Chaos	Fig. 11(j)
$[7.45, 8.3]$	Zero	Periodic orbit	Fig. 11(k)
$(8.3, 15]$	Positive	Chaos	Fig. 11(l)

(a) The periodic orbit for $b = 0.1$.(b) Multi period-doubling orbits for $b = 0.55$.(c) The complex period-doubling orbits for $b = 0.7$.(d) The chaos for $b = 0.85$, $LE = 1.496$.(e) The complex periodic orbit for $b = 1.6$.(f) The chaos for $b = 2.15$, $LE = 1.66$.Fig. 11. Different modes of system (1) versus b , when $a = 50$, $c = 13$, $d = 8$, $e = 33$, $f = 30$.

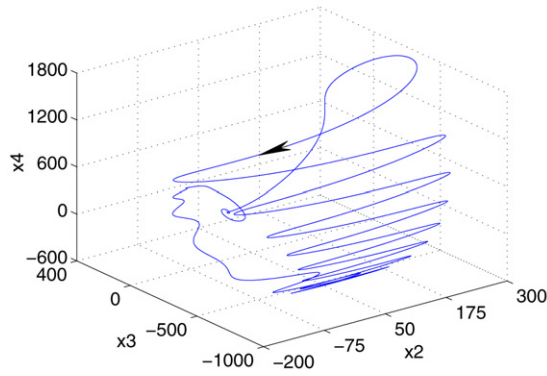
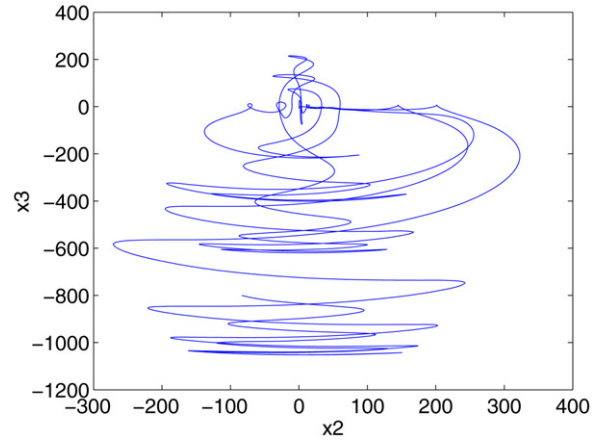
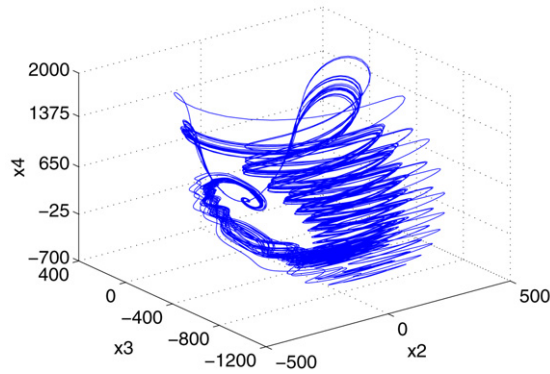
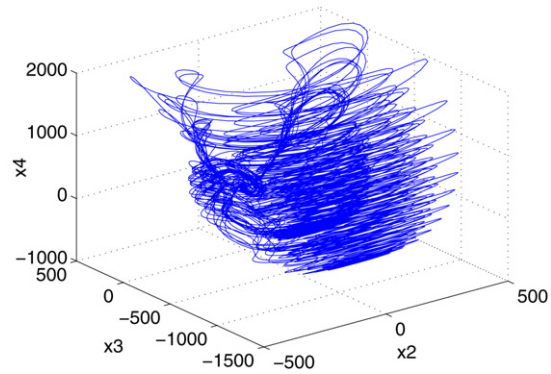
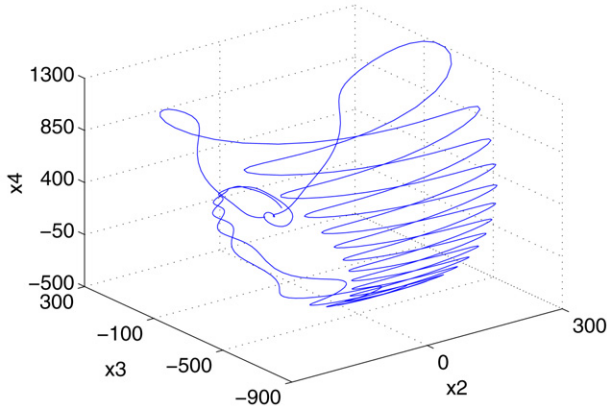
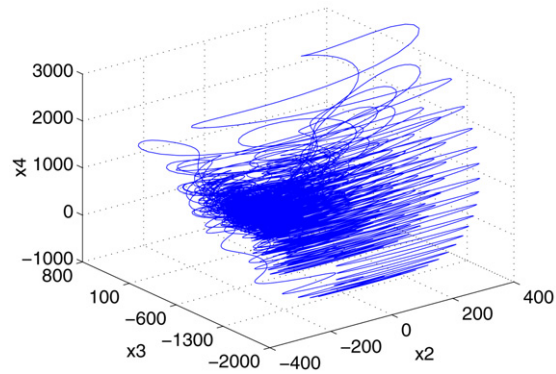
(g) The complex periodic orbit in 3D view for $b = 3$.(h) The projection on plane x_2 – x_3 of the complex periodic orbit for $b = 3$.(i) The chaos for $b = 4.025$, $LE = 1.771$.(j) The chaos for $b = 7.3$, $LE = 3.18$.(k) The complex periodic orbit for $b = 8$.(l) The chaos for $b = 13$, $LE = 6.44$.

Fig. 11. (continued)

extracted. Some hyperchaotic systems also have narrow high-magnitude spectral bandwidths. For example, Li et al. [16] presented a hyperchaotic system with positive LEs $l_1 = 0.5011$, $l_2 = 0.1858$, with the frequency spectrum shown in Fig. 12(b), where the bandwidth is less than 4.5 Hz.

Recently, we constructed a 3D chaotic system [20] with a large positive LE $l_1 = 5.2$. Its frequency spectrum bandwidth is about 13 Hz, as shown in Fig. 12(c). A new and more complicated four-wing chaotic system with a positive LE $l_2 = 6$ has also been generated by introducing a control term to merge two coexisting attractors of a 4D chaotic system [24,27] giving a prominent bandwidth of about 22 Hz, as

shown in Fig. 12(d). It is clear that the value of the leading positive LE plays an important role in determining the bandwidth and the degree of disorder of the orbits of a chaotic system.

When taking the parameters $a = 50$, $c = 26$, $d = 8$, $e = 33$, $f = 30$ in the new hyperchaotic system, there are two positive LEs, $l_1 = 13.4632$, and $l_2 = 3.478$. The frequency spectra of the signals x_1 and x_2 , generated numerically from the new hyperchaotic system, are shown in Figs. 13(a), (b), respectively. Note that the high-magnitude bandwidths of signals x_1 and x_2 are more than 95 Hz, which are more than 20 times wider than that of all the other systems mentioned above. One can

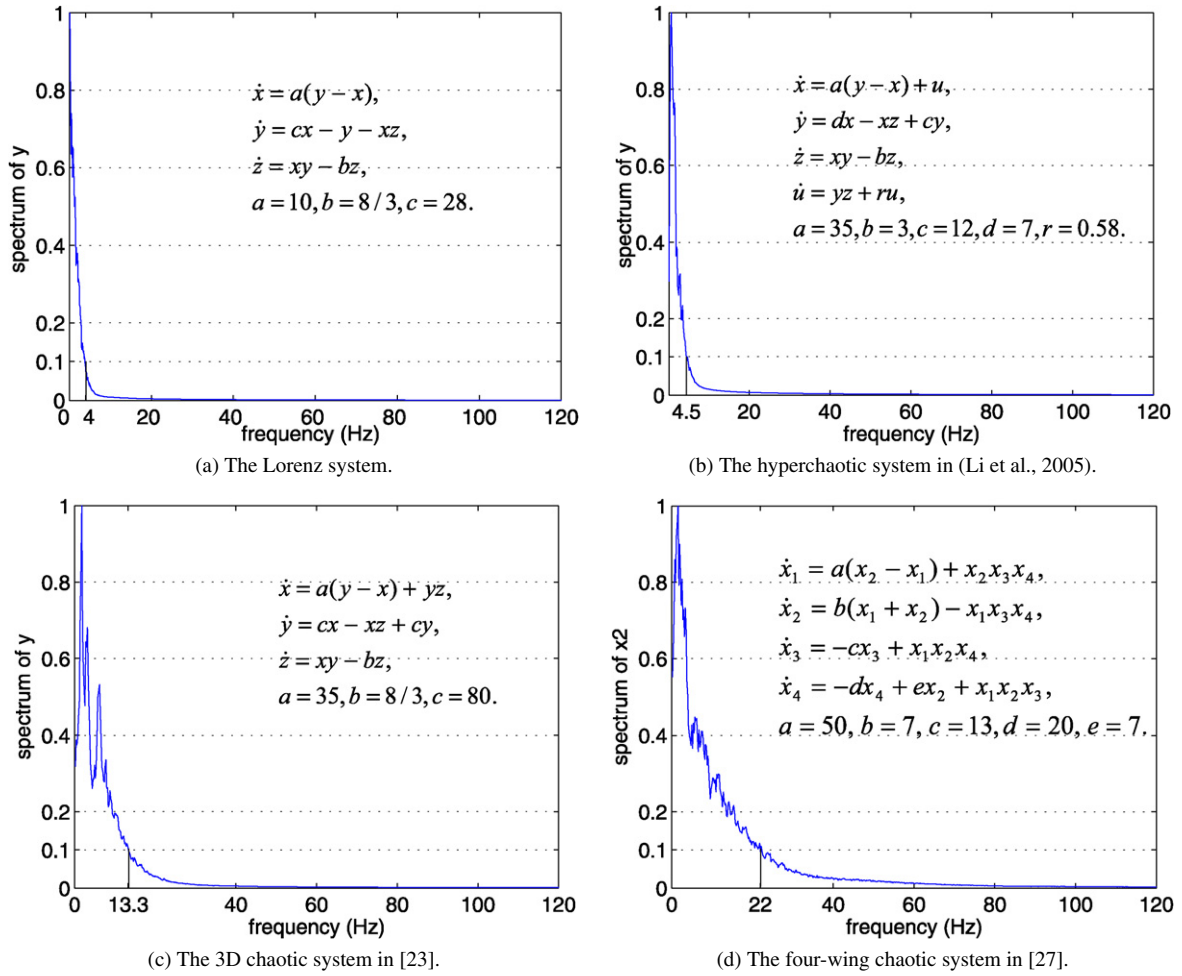
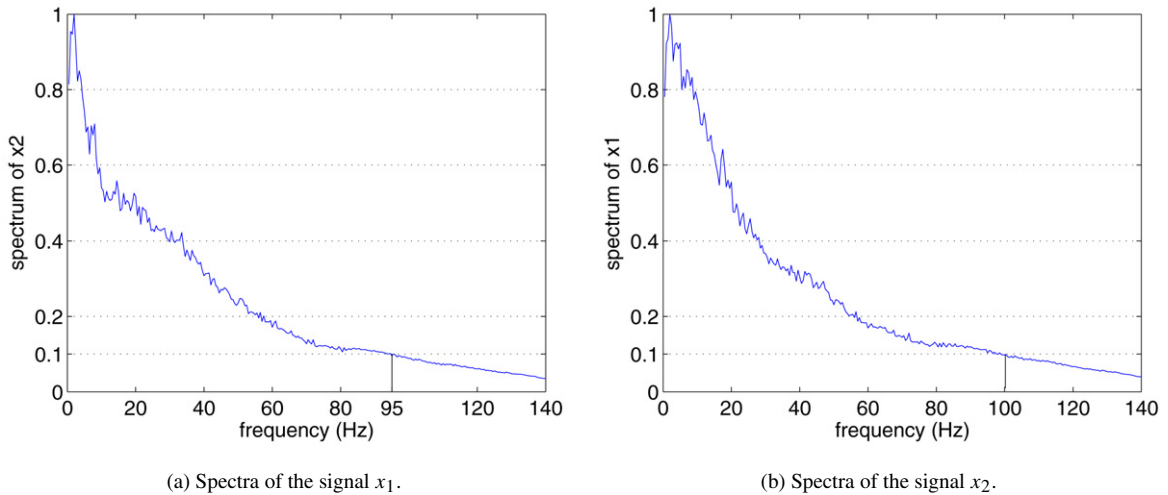


Fig. 12. Frequency spectra of three chaotic systems and the hyperchaos generated by Li et al. [30].

Fig. 13. Frequency spectra of the new hyperchaotic system, with $a = 42, b = 26, c = 13, d = 20, e = 50, f = 40$.

see that the frequency properties of the new hyperchaotic system are indeed exhibiting very strong randomness. In fact, the strong random property is also confirmed and demonstrated by the time response shown in Fig. 3, the Poincaré map shown in Fig. 8. If the frequency ranges with spectral values over 100^{-1} are considered. The bandwidths of the new hyperchaotic sig-

nals x_1 and x_2 are about 500 Hz. In sampling theory, an input signal (f_{in}) must be sampled with at least twice its frequency (Nyquist criteria, $f_{sample} \geq 2 \times f_{in}$). To keep the sample trajectory smooth enough, the sample frequency in practice can be 10 times or more than that of input, i.e. $f_{sample} = 10 \times f_{in}$. Therefore, the sample step should be $\tau = 1/(10 \times 500) = 2 \times 10^{-4}$

for the hyperchaotic system, which corresponds to our sample step mentioned in Section 3.1.

5. Conclusions

In this Letter, we have mathematically constructed built a new hyperchaotic system. Frequency spectra of this new hyperchaotic system, particularly its leading and second Les, show that the system is a truly hyperchaotic one within a large region of parameters, giving the system very strong randomness, high degree of disorder and extremely rich dynamics, as demonstrated by bifurcation analysis, Poincaré map analysis. More importantly, the frequency spectral analysis demonstrates that in the new system hyperchaos exists over an extremely broad high-magnitude bandwidth, which is very desirable for some engineering applications such as secure communications.

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